| OL | Temp | Hot | windy | Rain |
|---|---|---|---|---|
| Sunny | Hot | High | weak | No |
| Sunny | mild | High | Strong | NO |
| Cloudy | Hot | High | week | Yes |
| Rainy | mild | high | weak | yes |
| Rainy | ~~Cool~~ Normal | Normal | weak | yes |
| Rainy | Cool | Normal | strong | No |
| Cloudy | mild | Normal | strong | Yes |
| Sunny | Cool | Normal | weak | NO |

Target = Rain

Yes = 4 NO = 4

$= -\frac{4}{8}\log_2\frac{4}{8} - \frac{4}{8}\log_2\frac{4}{8}$

$= 2$

| Hot | Total | Yes | No |
|---|---|---|---|
| High | 4 | 2 | 2 |
| Normal | 4 | 2 | 2 |

$= -\frac{2}{4}\log_2\left(\frac{2}{4}\right) - \frac{2}{4}\log_2\left(\frac{2}{4}\right) = 1$

$-\frac{4}{8}\left(\frac{2}{4}\log_2\left(\frac{2}{4}\right) - \frac{2}{8}\log_2\left(\frac{2}{4}\log_2\left(\frac{2}{4}\right)\right.\right.$

$-\frac{4}{8}(1) - \frac{4}{8}(1)$

$= 0.3\,4\,4$

outlook = 0.65

Temp = 0.61

H = 0

windy = 0.04

∴ high = outlook

weight  HS-weightage
= 0.655

| wind | Total | Yes | No |
|---|---|---|---|
| weak | 5 | 3 | 2 |
| Strong | 3 | 2 | 2 |

w.e 5/8 (.971) + 3/8 (0.918)

IG = 1.00 − 0.951
= 0.049

| Temp | Total | Yes | No |
|---|---|---|---|
| Hot | 2 | 1 | 1 |
| mild | 3 | 2 | 1 |
| Cool | 3 | 1 | 2 |

WE = 3/8 (.1) + 3/8 (.918) + 2/8 0.9

IG = 1.00 − 938
= 0.615